Intel. An example of this is that the Pentium D 820, which is a dual-core processor, runs as a single-core on these boards.

Though all the boards supported DDR2, except for the MSI 915G Neo3-FIR, which supported DDR 400MHz, the speeds varied. Some boards supported DDR2 at 533 MHz, while others supported 667 MHz. The ASUS P5AD2-E Premium supported an odd 711 MHz, while the Gigabyte i-DNA GA-8I955X Royal supported an equally odd 888 MHz.

Higher memory speeds are advantageous when you are into multimedia applications and gaming, where that little extra bandwidth can make a lot of difference.

The motherboards based on Intel 945G and 955X as well as those based on nVidia's nForce4 SLI supported the faster and emerging SATA II standard, while being backward compatible with SATA I drives. The ASUS P5AD2-E Premium, P5ND2-E Deluxe and Foxconn 955X7AA sported eight SATA ports, which allow you to add up to eight SATA devices—which is great from the upgradeability point of view.

The ASUS Proactive P5AD2-E Premium Wireless Edition board had IEEE 802.11g-compliant Wi-Fi on-board, which allows data transmission of up to 54 Mbps. The MSI P4N

Asus Proactive P5ND2-SLI Delux

Diamond goes even further by bundling a Bluetooth and IEEE 802.11g combo PCI card which, if purchased separately, retails at Rs 2,900. If you have a Bluetooth-enabled phone, it is very convenient to transfer files from the phone to the PC. The Gigabyte i-DNA GA-8I955X Royal also shipped with a Bluetooth USB dongle—you just plug in the dongle to a USB port, install the drivers, and your PC is Bluetooth-enabled.

An external SATA port was present on the ASUS P5ND2-SLI Deluxe, known as SATA-On-The-Go, which has hot-plug support. Up to 16 devices can be connected with port multiplier functions.

SPDIF-Out was a feature common on all the boards, with some even providing SPDIF-In and coaxial Out. The Intel 945G-based boards also had integrated Intel graphics, though if you're purchasing this board, you're expected to purchase a separate PCIe graphics card to fully utilise the power of the system.

The Gigabyte i-DNA GA-8I955X Royal came with a U-Plus DPS (Dual Power System), which is a revolutionary circuit built to withstand varying AC power levels for system protection and stability.

Several boards featured excellent audio solutions, one of which was the Intel HDA (high-definition audio), which produces high-fidelity audio output. The MSI P4N Diamond had a hardware Creative SoundBlaster Live! 24-bit chip as the audio solution. (Most onboard sound solutions involve the CPU; this 24-bit chip takes the load off the CPU.) This is no doubt one of the best on-board audio solutions.

The Gigabyte i-DNA GA-8I955X Royal sported dual BIOS, where there's a backup BIOS on the motherboard. If the main BIOS gets corrupted, the backup BIOS takes over and the main BIOS can be flashed to restore it. The MSI motherboards also featured similar functionality. ECS featured Top-Hat Flash. (Refer the AMD section for more on Top-Hat Flash.)

In addition to this, all the boards had overclocking of some sort. MSI featured the 'Cell menu' in the BIOS, while ASUS has 'AI overclocking' and 'non-delay overclocking'. In all these cases, there were options to tweak the FSB, memory frequency, timings and CPU core and memory voltages in order to overclock the system.

drives to duplicate data. Striping is not involved here. This offers the best ratio of performance to fault tolerance.

RAID 0+1: This combines the best features of RAID 0 and 1. In this type of RAID, two pairs of striped drives are mirrored together to provide fault tolerance. Thus, it requires four hard drives to implement.

SATA: Also abbreviated as S-ATA, is short for Serial ATA, which is an interface for physical storage devices such as hard drives. It is a serial link and consists of a single cable with a minimum of four wires creating a point-to-point connection between devices. The transfer rates for SATA begin at 150 MBps, and for SATA II it is 300 MBps.

SLI: Short for Scalable Link Interface, which is a graphics standard in which the motherboard has two PCIe graphics slots so that two cards can be plugged in simultaneously and connected via a bridge. This setup offers a huge performance leap in graphics applications such as gaming.

Southbridge: This is a chip on the motherboard that controls all the onboard devices including the IDE bus and the PCI bus. This is the second-largest chip on the motherboard, and is sometimes covered by a heat sink.

Wi-Fi: An abbreviation for Wireless Fidelity, Wi-Fi is a wireless radio standard that was developed for communication between mobile devices such as laptops, PDAs, and LANs, but is now even being used in desktop computers.